

## HW2 Q3: Capacity Allocation after Poisson Splitting

IEDA 1250 Queueing Theory

**Can we treat the system as two independent M/M/1 queues?**

Yes. By Poisson splitting, the two arrival streams are Poisson with rates

$$\lambda_1 = p\lambda, \quad \lambda_2 = (1-p)\lambda.$$

So we can model the two branches as two M/M/1 queues, but the overall mean response time must be weighted by the routing probabilities:

$$E[T] = p \frac{1}{\mu_1 - p\lambda} + (1-p) \frac{1}{\mu_2 - (1-p)\lambda}, \quad \mu_1 + \mu_2 = \mu.$$

**Stability conditions**

$$\mu_1 > p\lambda, \quad \mu_2 > (1-p)\lambda.$$

These are the minimum capacities needed just to keep both queues stable.

**Easy case:**  $p = \frac{1}{2}$ .

By symmetry, split capacity equally:

$$\mu_1^* = \mu_2^* = \frac{\mu}{2}.$$

Each queue has arrival rate  $\lambda/2$ , so

$$E[T_1] = E[T_2] = \frac{1}{\mu/2 - \lambda/2} = \frac{2}{\mu - \lambda}.$$

Thus

$$E[T] = \frac{2}{\mu - \lambda}.$$

**General case:**  $p > \frac{1}{2}$ .

First allocate the minimum stable capacities:

$$p\lambda \quad \text{to server 1,} \quad (1-p)\lambda \quad \text{to server 2.}$$

The remaining extra capacity is

$$\mu - \lambda.$$

Define the extra capacities

$$x_1 = \mu_1 - p\lambda, \quad x_2 = \mu_2 - (1-p)\lambda.$$

Then

$$x_1 + x_2 = \mu - \lambda, \quad E[T] = \frac{p}{x_1} + \frac{1-p}{x_2}.$$

Minimizing this gives

$$\frac{x_1}{x_2} = \frac{\sqrt{p}}{\sqrt{1-p}}.$$

**Optimal fraction of extra capacity going to server 1**

$$\frac{x_1}{x_1 + x_2} = \frac{\sqrt{p}}{\sqrt{p} + \sqrt{1-p}}$$

Therefore, the optimal capacity allocation is

$$\mu_1^* = p\lambda + (\mu - \lambda) \frac{\sqrt{p}}{\sqrt{p} + \sqrt{1-p}}$$

$$\mu_2^* = (1-p)\lambda + (\mu - \lambda) \frac{\sqrt{1-p}}{\sqrt{p} + \sqrt{1-p}}$$

**Intuition:** each server first receives enough capacity to handle its own arrival rate. The remaining capacity should be split in proportion to  $\sqrt{p}$  and  $\sqrt{1-p}$ , not directly in proportion to  $p$  and  $1-p$ .